PROJECT REPORT

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AI SKIN INTELLIGENCE & PERSONALIZED SKINCARE PLATFORM

1. Introduction

The AI Skin Intelligence & Personalised Skincare Platform is an advanced skincare system that connects everyday skincare with professional dermatological care. It helps people understand their skin, choose suitable products, and get guidance from skincare professionals and dermatologists.

The skin is the human body's largest organ, serving as a complex biological barrier against pathogens, ultraviolet radiation, and environmental pollutants. Maintaining optimal skin barrier health requires an understanding of diverse physiological and lifestyle factors, including genetic skin type, epidermal hydration, sleep architecture, chronic stress, environmental pollution, and product chemical compatibility.

Traditionally, consumers select skincare regimens through unguided trial-and-error, word-of-mouth suggestions, or marketing campaigns. These methods frequently lead to adverse reactions, contact dermatitis, and damaged skin barriers. Conversely, medical professionals lack standardized tools to monitor patient daily routine compliance and track recovery between appointments.

2. Problem Statement

Modern skincare users and dermatological professionals face four major challenges.

i. **Difficulty in Selecting Suitable Skincare Products:** Many consumers struggle to identify products that match their skin type, concerns, and sensitivities. Repeatedly purchasing and testing unsuitable products can waste money and worsen existing skin problems.

ii. **Unsafe Ingredients and Incompatible Product Combinations:** Skincare products may contain active ingredients such as retinoids, L-ascorbic acid (vitamin C), and alpha- or beta-hydroxy acids (AHAs/BHAs). Incorrect use or unsuitable combinations can cause irritation, damage the skin's protective barrier, and contribute to pigmentation changes.

iii. **Limited Professional Oversight in Digital Skincare Applications:** Many existing skincare applications focus primarily on product promotion and basic questionnaires. They may lack transparent skin health assessments, comprehensive ingredient safety checks, and appropriate professional dermatological supervision.

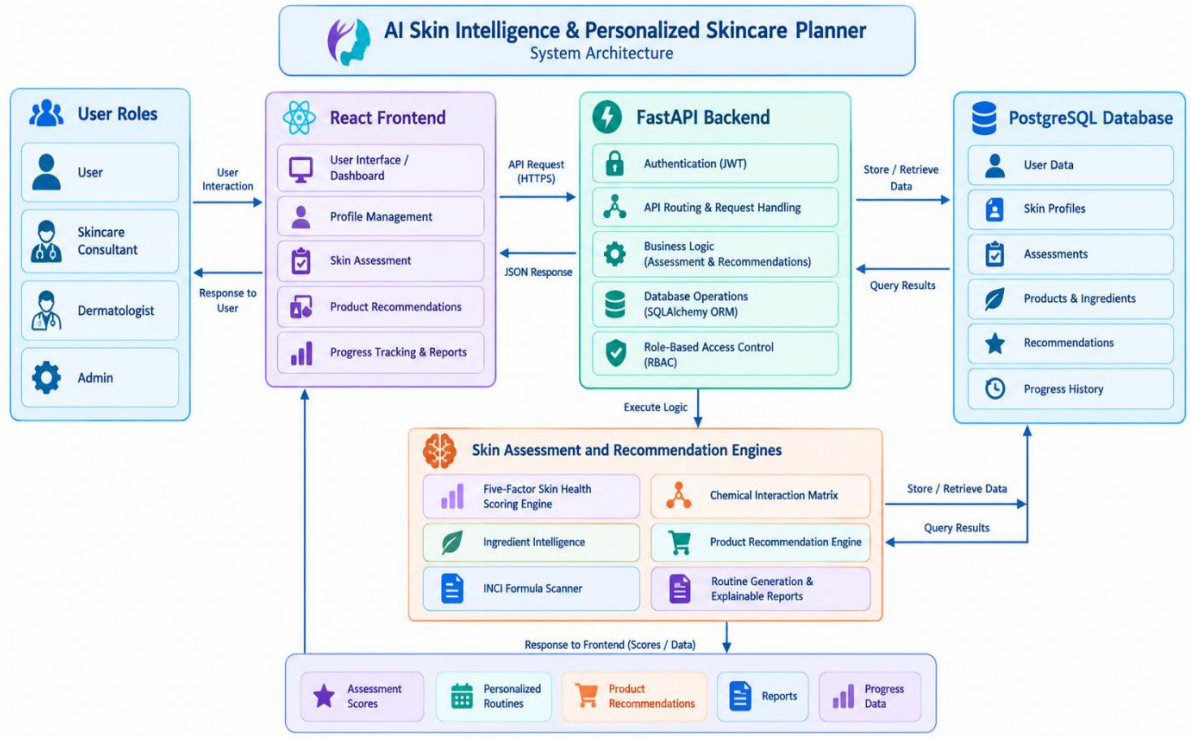
iv. **Challenges in Tracking Skincare Habits and Treatment Progress:** Dermatologists and skincare consultants may lack a unified system to monitor patients' daily routine adherence, track changes in skin health over time, and update skincare plans between consultations.

3. Proposed Solution

The **AI Skin Intelligence Platform** provides an integrated digital ecosystem that combines consumer skincare with clinical decision support:

- **12-Factor Biometric & Lifestyle Questionnaire:** Evaluates skin type, age group, concerns, allergens, sensitivities, sleep duration/quality, stress, physical activity, water intake, and environmental exposure.
- **Explainable Hybrid Skin Health Engine:** Combines a clinical baseline heuristic (70%) with a trained XGBoost Regressor (30%) to generate an objective barrier score (0–100) with transparent category-level deductions.
- **Safety-First Product Recommendation Engine:** Matches user profiles against product formulations using multi-hot vectorization and Cosine Similarity, automatically filtering out declared allergens, synthetic fragrances, and drying alcohols.
- **Atmospheric Climate Sync:** Automatically adjusts morning and evening regimens based on live meteorological data (e.g., mandating high-SPF filters during high UV index or prescribing occlusives in low-humidity environments).
- **Four Interconnected Role Portals:** Dedicated, token-guarded interfaces for Users, Skincare Consultants, Dermatologists, and Administrators.
- **Clinical Triage & Digital Prescriptions:** Enables certified dermatologists to review patient charts, triage barrier urgency, write digital prescriptions, and compile formal PDF Clinical Dossiers.
- **Production-Grade Cloud Deployment:** Packaged into three synchronized Docker containers (frontend, backend, db) running on an AWS EC2 instance with persistent disk volumes.

4. System Architecture



Frontend (Presentation Layer)

i. ***Primary Technologies:*** React 19, Vite, React Router DOM, and Axios.

ii. ***Role in the System:***
The frontend is the interactive visual layer running inside the user's browser. It is built as a Single Page Application (SPA), meaning pages transition smoothly without full browser reloads.

iii. ***Key Responsibilities:***

- Role-Specific Dashboards: Delivers custom interfaces depending on login credentials—a habit-tracking hub for Users, an adherence monitor for Consultants, a triage queue and prescription pad for Dermatologists, and a telemetry dashboard for Admins.
- Data Collection Forms: Collects the 12-factor biometric profile (skin type, lifestyle, sleep, water intake, allergies, sensitivities) through structured questionnaires.
- Dynamic Host Interceptor: Includes an Axios request interceptor (main.jsx) that automatically detects the browser's hostname. When accessed on the AWS EC2 cloud, it redirects all API calls from localhost to the cloud server IP on port 5000, ensuring remote users can communicate with the backend seamlessly.

- Visual Feedback & Reports: Renders 30-day health score trajectory charts, recovery percentage badges, daily routine checklists, and provides one-click triggers to download clinical PDF dossiers.

Backend (Application & API Gateway Layer)

i. ***Primary Technologies:*** Python, Flask 3, Flask-JWT-Extended, Werkzeug, and Flask-CORS.

ii. ***Role in the System:***
The backend acts as the central conductor of the platform. It receives HTTP requests from the React frontend, validates security tokens, executes business logic, communicates with the database, and orchestrates machine learning workflows.

iii. ***Key Responsibilities:***

- Stateless Authentication & RBAC: When users log in, the backend verifies their credentials using salted PBKDF2 SHA-256 password hashing. It issues cryptographically signed JSON Web Tokens (JWT) with embedded role claims. Every sensitive endpoint enforces @jwt_required() and checks user roles, blocking unauthorized access with 401 Unauthorized or 403 Forbidden.
- Modular Blueprint Routing: Code is cleanly separated into Blueprints:
  - main: Handles user accounts, profile CRUD operations, daily checklists, clinical doctor prescriptions, and admin user controls.
  - reports_bp: Dedicated service managing high-definition document exports.
- Document Generation Pipelines: Uses ReportLab to build multi-page vector PDF Clinical Dossiers and OpenPyXL to assemble multi-sheet Excel workbooks dynamically in memory, streaming them directly to the client as downloadable binary blobs.

Database (Data Persistence Layer)

i. ***Primary Technologies:*** PostgreSQL 15 managed through Flask-SQLAlchemy ORM.

ii. ***Role in the System:***
The database provides structured, transactional, ACID-compliant relational storage. It ensures that all patient histories, doctor notes, and product catalogs remain permanent and secure.

iii. ***Key Responsibilities:***

- Normalized Relational Schema (7 Core Tables):
    1. users: Stores user credentials, salted passwords, and system roles (user, consultant, dermatologist, admin).
    2. skin_profile: Stores 12-factor biometric, physiological, and environmental diagnostics.
    3. score_history: Records chronological barrier health scores and assessment dates.
    4. daily_checklist: Maintains daily morning and evening tasks with completion booleans.
    5. prescriptions: Stores formal medical treatments written by dermatologists (medication, dosage, frequency, course duration).
    6. clinical_recommendations: Stores physician advisory notes and routine modifications.
    7. products & ingredients: Manages cosmetic formulations, active ingredients, comedogenic ratings, and price tiers.
- Security & Injection Immunity: All database interactions use SQLAlchemy parameterized queries, preventing SQL injection vulnerabilities.
- Cloud Volume Persistence: Containerized using Docker with a named volume (pgdata). This guarantees that all patient data, accounts, and clinical records are saved directly to the cloud server disk and survive container updates or system reboots.

Machine Learning & Intelligence Layer

i. ***Primary Technologies:*** XGBoost Regressor, Scikit-Learn, NumPy, Pandas, and Open-Meteo Weather API.

ii. ***Role in the System:***
This layer serves as the computational brain, transforming raw biometric and lifestyle data into explainable scores, safe product recommendations, and adaptive routines.

iii. ***Key Responsibilities:***

- Hybrid Skin Barrier Health Scoring Engine:
  Calculates an objective barrier health score (0–100) using an ensemble

approach: Final Score $=$ round$(0.70 \times$ Clinical Baseline $+ 0.30 \times$ XGBoost Predictor)

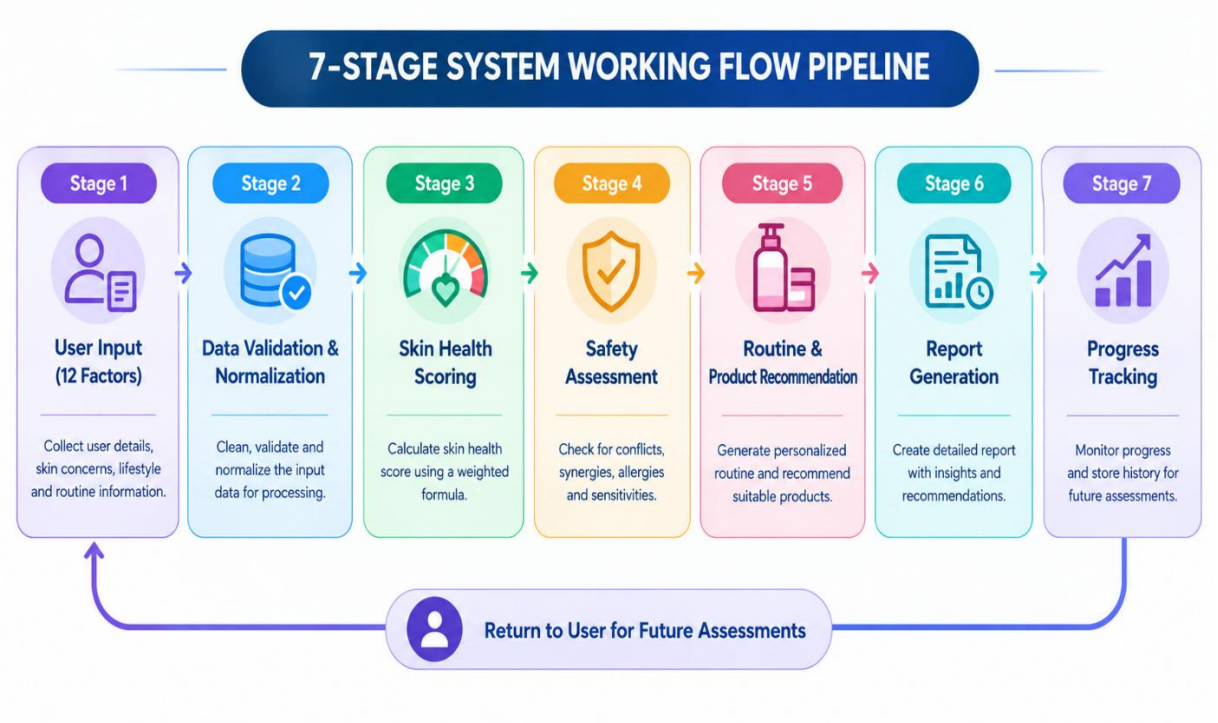
1. *The 70% Clinical Baseline:* A transparent heuristic evaluating Skin Condition (35%), Lifestyle (20%), Sleep (15%), Adherence (20%), and Hydration (10%).
2. *The 30% XGBoost Regressor:* Trained on multi-dimensional patient vectors ($R^2 = 0.912$, RMSE $= 3.24$) to capture complex, non-linear interactions (e.g., how chronic stress multiplies the negative impact of poor sleep on barrier repair).

- Content-Based Product Recommender:
  Encodes product attributes and user profiles into an L_2-normalized vector space using Scikit-Learn's MultiLabelBinarizer. It computes pairwise Cosine Similarity to match and rank safe cleansers, serums, moisturizers, and sunscreens by compatibility (55%–98%).
- Chemical Safety & Conflict Screener:
  Enforces hard exclusions before running recommendations: any product containing declared patient allergens or sensitizing agents (fragrance, alcohol) is completely excluded. It also screens for pairwise chemical clashes (e.g., separating Retinoids from direct AHAs to prevent over-exfoliation).
- Atmospheric Climate Sync:
  Queries the Open-Meteo API for live environmental metrics (UV index, humidity, ambient temperature) based on location, dynamically enforcing high SPF during high UV exposure or recommending rich ceramide occlusives during low-humidity weather.

5. Technology Stack

Layer	Technologies	Purpose
Frontend	React, Vite, React Router DOM, Axios	User interface, navigation, and API communication.
Backend API	Python, Flask, Flask-JWT-Extended, Werkzeug, Flask-CORS	REST API, authentication, password security, and cross-origin access.
Database	PostgreSQL, Flask-SQLAlchemy	Relational data storage and database operations.
AI & Machine Learning	XGBoost, Scikit-learn, Pandas, NumPy	Predictive modelling, similarity-based recommendations, and data processing.
Document Export	ReportLab, OpenPyXL	PDF report generation and Excel export.
External API	Open-Meteo API	Environmental data, including UV index, humidity, and temperature.
DevOps & Deployment	Docker Compose, AWS EC2, Ubuntu 22.04	Container orchestration and cloud hosting.

6. Methodology and Workflow



The platform executes a sequential, 7-stage data pipeline that converts raw user inputs into personalised clinical guidance:

Stage 1: User Input

Collects multi-dimensional inputs across skin type, concerns, lifestyle, sleep, hydration, existing routines, environmental exposure, and reported allergies/sensitivities.

Stage 2: Data Validation

Sanitizes incoming JSON payloads, handles missing values, normalizes numerical metrics, and securely upserts the structured profile into the PostgreSQL database.

Stage 3: Skin Health Scoring

Computes the objective barrier health score (0–100) using the five-factor model (Skin Condition 35%, Lifestyle 20%, Sleep 15%, Adherence 20%, Hydration 10%) combined with the machine learning regressor.

Stage 4: Safety Assessment

Conducts zero-tolerance allergen screening, sensitivity trigger identification, pairwise active chemical clash detection, and risk factor extraction.

Stage 5: Routine & Product Recommendations

Generates customized morning and evening routines while using vectorized Cosine Similarity to recommend safe, compatible products.

Stage 6: Report Generation

Compiles factor-level breakdowns, diagnostic metrics, and routine plans into on-the-fly downloadable PDF and Excel formats.

Stage 7: Progress Tracking

Analyzes score trends over time, records routine adherence percentages, tracks biomarker recovery, and loops back (Return to User) for iterative reassessments.

7. Modules and Functionalities

User Authentication & Role-Based Access Control (RBAC)

i. **Purpose:** Ensures secure platform onboarding, identity verification, and role segregation.

ii. **Functionality:**

- ***Secure Registration & Login:*** Accepts user credentials (POST /register, POST /login) and hashes passwords using salted PBKDF2 SHA-256 encryption via Werkzeug.
- ***Stateless JWT Tokens:*** Issues cryptographically signed JSON Web Tokens (30-day validity) containing identity and role claims.
- ***Role Segregation***: Enforces permissions across four dedicated roles:
    1. user: Accesses personal diagnostics, checklists, and product recommendations.
    2. consultant: Manages client profiles, monitors compliance, and adjusts aesthetic routines.
    3. dermatologist: Triages clinical patient queues, reviews electronic medical records, writes digital prescriptions, and downloads medical dossiers.
    4. admin: Governs system telemetry, reassigns user roles, and performs master database exports.
- ***Route Protection:*** Server-side @jwt_required() decorators and client-side route guards reject unauthorized access with 401 Unauthorized or 403 Forbidden.

Skin Profile Management

i. **Purpose:** Establishes a permanent, structured biological and lifestyle baseline for every user.

ii. **Functionality:**

- ***12-Factor Data Capture:*** Captures skin type (oily, dry, combination, sensitive, normal), age bracket, skin concerns, chemical allergies, sensitivities, sleep duration, sleep quality, stress levels, physical exercise, qualitative hydration, daily water intake (Liters), and environmental exposure (sun/pollution).

- ***Profile CRUD Operations:*** Managed through GET and POST /profile, enabling users to retrieve and update their parameters as lifestyle habits evolve.
- ***Database Synchronisation:*** Automatically links the profile one-to-one with the user's authentication account in PostgreSQL with timestamped audit updates.

Skin Assessment Engine

i. **Purpose:** Analyzes raw profile inputs to extract primary clinical complaints, calculate risk factors, and prioritize treatments.

ii. **Functionality:**

- ***Concern Identification & Severity Ranking:*** Evaluates multiple declared complaints (e.g., active acne, hyperpigmentation, fine lines, redness) and ranks them using clinical severity heuristics (prioritizing acute inflammatory conditions over general cosmetic aging).
- ***Risk Factor Analysis:*** Detects underlying physiological triggers (e.g., *"High stress combined with poor sleep amplifies inflammatory acne and slows cellular barrier repair"*).
- ***Fitzpatrick Barrier Diagnostic:*** Evaluates skin barrier vulnerability based on reported reactivity and sensitivity markers.

Personalized Routine Generator

i. **Purpose:** Constructs structured, step-by-step morning, evening, and weekly treatment roadmaps.

ii. **Functionality:**

- ***AM Regimen (Protect & Hydrate):*** Cleanser →Antioxidant/Treatment Serum →Lightweight Barrier Moisturizer →Broad-Spectrum Sun Protection (SPF 50+).
- ***PM Regimen (Repair & Regenerate):*** Gentle Double Cleanser →Targeted Active Treatment →Nourishing Lipid/Ceramide Night Cream.
- ***Weekly Treatment Planning:*** Assigns specific active nights (e.g., Tuesday for gentle chemical exfoliation, Wednesday/Friday for Retinoids, Sunday for barrier hydration masks).

- ***Dynamic Weather Adaptation:*** Integrates with the Open-Meteo API to fetch real-time UV index, humidity, and temperature, dynamically enforcing high SPF during high UV days or prescribing occlusives in dry, low-humidity climates.

Ingredient Intelligence Module

i. **Purpose:** Provides chemical education, screens formulations for safety, and prevents harmful active combinations.

ii. **Functionality**:

- ***8 Core Active Ingredient Families:*** Comprehensive knowledge base detailing *Retinoids, Niacinamide, Vitamin C, Hyaluronic Acid, Salicylic Acid (BHA), Ceramides, Peptides, and AHAs (Glycolic/Lactic Acid)*.
- ***Chemical Interaction Matrix:*** Identifies pairwise conflicts (e.g., separating Retinol from direct exfoliating acids to prevent barrier stripping) and highlights synergy pairs (e.g., Vitamin C + Sunscreen for photoprotection).
- ***Allergy & Sensitivity Screener:*** Automatically flags products containing declared user allergens, drying alcohols, or synthetic fragrances.
- ***INCI Formula Matching***: Standardizes chemical names against International Nomenclature of Cosmetic Ingredients (INCI) standards.

Product Recommendation Engine

i. **Purpose:** Matches the user's specific skin profile with safe, effective, and budget-friendly commercial skincare products.

ii. **Functionality:**

- ***Content-Based Filtering (Cosine Similarity):*** Uses Scikit-Learn's MultiLabelBinarizer to encode product attributes into an L_2-normalized vector space, computing pairwise Cosine Similarity against the user's preference vector (55%–98% compatibility match).
- ***Hard Clinical Exclusions:*** Completely eliminates any product containing declared user allergens before ranking similarity.
- ***Catalog Categorization:*** Organizes products across 7 categories: *Face Wash, Toners, Serums, Moisturizers, Sunscreens, Treatments, and Face Masks*.

- ***Budget & Price Tiering:*** Sorts recommendations into *Budget*, *Mid-tier*, and *Premium* clinical tiers with real pricing in INR.

Skin Health Scoring Engine

i. **Purpose:** Calculates an objective, repeatable skin barrier health score (0 to 100) to track skin wellness.

ii. **Functionality:**

- ***Five-Factor Clinical Baseline (70% Weight):***

  $$\text{Baseline} = 0.35S + 0.20L + 0.15R + 0.20A + 0.10H$$

  1. Baseline Skin Condition (35%)
  2. Lifestyle Habits & Stress (20%)
  3. Sleep & Recovery Quality (15%)
  4. Routine Consistency & Adherence (20%)
  5. Hydration Level (10%)

- ***XGBoost Machine Learning Regressor(30% Weight):***
  Trained on 17 biometric input features ($R^2 = 0.912$, RMSE $= 3.24$) to capture complex, non-linear interactions across compounding lifestyle stressors.

- ***Ensemble Score:*** Merges clinical rules and machine learning into an explainable numerical score with level badges: Critical (≤ 40), Fair (41–55), Average (56–70), and Good (71–100).

Progress Tracking & Analytics

i. **Purpose:** Quantifies skin recovery, evaluates routine consistency, and visualises longitudinal improvement.

ii. **Functionality:**

- ***Rolling 7-Day Adherence Tracking***: Calculates routine compliance from completed checklist items: $\text{Adherence Rate} = \left(\frac{\text{Completed Daily Steps}}{\text{Total Assigned Daily Steps}}\right) \times 100$

- ***Longitudinal Score Trajectory:*** Graphs 30-day health score changes to illustrate progress trends (Improving, Stable, or Declining).

- ***Biomarker Recovery Cards:*** Tracks percentage improvements across four key skin indicators:
    1. Erythema / Redness Reduction (%)
    2. Skin Texture Smoothness (%)
    3. Spot & Hyperpigmentation Fading (%)
    4. Stratum Corneum Hydration Gain (%)

Dashboards & Analytics (Across All 4 Roles)

i. **Purpose:** Delivers dedicated, role-specific workspaces tailored to each stakeholder's responsibilities.

ii. **Functionality:**

- ***User Dashboard***: Displays overall skin health score, streak counter, daily AM/PM checklist, ML-matched products, and recovery trajectory.
- ***Consultant Dashboard***: Searchable client directory, client lifestyle evaluation, and color-coded adherence indicators (Consistent $\geq 80\%$vs. Needs Follow-up $< 80\%$).
- ***Dermatologist Dashboard*** : Clinical patient triage queue prioritized by barrier urgency (Critical ≤ 40), electronic medical chart viewer, digital prescription pad, clinical treatment notes, and one-click Clinical Dossier PDF export.
- ***Admin Dashboard****:* Real-time telemetry counters (total users, completed profiles, health scores, prescriptions, catalog items), user directory with live role reassignment, system event audit log, and full database Excel backups.

Notification & Reminder System

i. **Purpose:** Supports patient compliance, encourages habit building, and prevents treatment abandonment.

ii. **Functionality:**

- ***Routine Reminders***: Prompts users to complete their morning sun protection and evening repair steps.
- ***Hydration Alerts:*** Notifies users falling behind on daily water intake targets based on their profile.

- ***Product Replenishment Alerts:*** Calculates estimated product depletion timelines based on daily checklist logging (alerting users when a 50ml serum or sunscreen is nearing empty).
- ***Clinical Follow-up Alerts:*** Flags clients with low consistency ($< 80\%$) on consultant and doctor dashboards to prompt clinical check-ins.

Reports & Export System

i. **Purpose:** Provides on-demand generation of formal, high-definition clinical documentation and database backups.

ii. Functionality:

- ReportLab Vector PDF Compilation:
  1. *Skin Assessment Diagnostic Report:* Complete breakdown of the 5-factor model and identified risk triggers.
  2. *Personalized Routine Plan:* Step-by-step AM/PM and weekly active treatment roadmaps.
  3. *Product Recommendations Report:* Matched product formulations, suitability percentages, and ingredient safety notices.
  4. *Skin Health Progress Report:* 30-day historical score logs, adherence rates, and progress velocity.
  5. *Patient Clinical Dossier:* Formal medical document combining patient demographics, barrier scores, active prescriptions, and physician notes.
  6. *Platform Master Audit Report:* Executive administrative audit of platform health, security, and user distribution.
- OpenPyXL Multi-Sheet Excel Export: Compiles all relational database tables into a structured 5-sheet workbook (Users, Skin Profiles, Health Scores, Prescriptions, Product Catalog) for offline analysis and clinical backup.

8. Project Scope and Future Enhancements

Project Scope & Operational Boundaries

- **Clinical Boundary:** The platform serves as a personalized skincare recommendation and clinical decision-support system. It does not replace emergency medical interventions or formal in-person diagnostic biopsies.
- **Data Security & Privacy:** All passwords use PBKDF2 hashing, endpoints enforce JWT verification, and database interactions use parameterized ORM queries to prevent SQL injection.
- **Production Deployment:** Designed for containerized deployment via Docker Compose on AWS EC2, with data persistence managed through named Docker volumes.

Future Enhancements

1. **Computer Vision Skin Lesion Diagnostics:**
   Integrate deep convolutional neural networks (e.g., EfficientNet or ResNet-50) to analyse user-submitted skin photos for automated classification of acne lesions, erythema, and dark spots.
2. **Wearable IoT Biometric Integration:**
   Connect with smart health platforms (Apple Health, Google Health Connect) to automatically sync daily sleep duration, step counts, and stress metrics.
3. **Automated Product Replenishment:**
   Implement inventory tracking that calculates product usage based on checklist completion and alerts users when formulations are nearing depletion.
4. **Mobile Camera OCR for Ingredient Labels:**
   Incorporate Optical Character Recognition (OCR) on mobile devices to let users scan physical product ingredient lists in stores for instant compatibility and allergy checks.
5. **Multi-Language Localization:**
   Expand the interface and PDF reports to support multiple global languages for international clinical use.

9. Conclusion

The **AI Skin Intelligence & Personalized Skincare Platform** was designed, developed, and deployed to address the challenges of trial-and-error skincare, misleading social media trends, and marketing claims that can lead to skin irritation, barrier damage, and unnecessary expenses. By integrating full-stack web technologies, explainable AI/ML techniques, ingredient safety analysis, and professional dermatological oversight, the platform delivers personalized, data-driven skincare recommendations while supporting informed decision-making. All planned project milestones were implemented, validated, and deployed in a live production environment, establishing an integrated system that connects daily skincare management with professional clinical guidance. Through personalized routines, systematic ingredient screening, progress tracking, and support for professional consultations and clinical documentation, the platform promotes a safer, more transparent, and proactive approach to skincare, demonstrating how intelligent automation and clinical expertise can work together to improve digital dermatological wellness.